MATH 423: Lecture 6

Portfolio Management
Capital Asset Pricing Model

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Contents

(1) Beta factor & CAPM
(2) Security market line
(3) Value at risk (VaR)

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Given a portfolio V, we want to understand how the return K_V will react to trends affecting the whole market, i.e., the return K_M of the market portfolio.

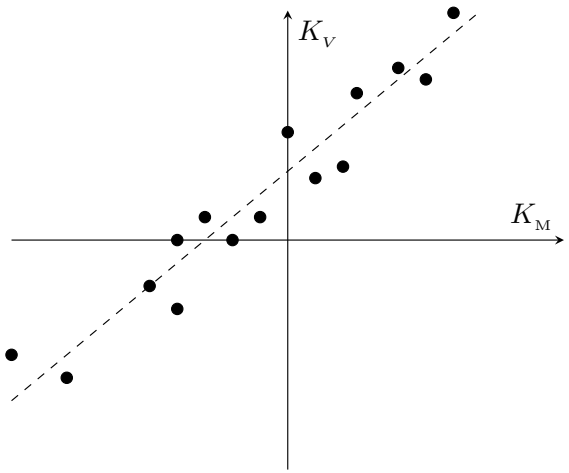
Definition 6.1

The beta factor of a portfolio/security V is defined by $\beta_V = \dfrac{Cov(K_V, K_M)}{\sigma_M^2}$.

For each scenario ω, we plot the pair $\big(K_M(\omega), K_V(\omega)\big)$ onto a plane;

In general, all points are randomly scattered, so will not lie on a same line;

Give a line on the plane $y = \beta x + \alpha$ with slope β and vertical intercept α, we see that the r.v. $|K_V - (\beta K_M + \alpha)|$ describes the vertical distance of each point $\big(K_M, K_V\big)$ to the line;



We want to find a best line, or equivalently a pair of (β, α), that minimizes

$$\mathbb{E}\Big[\big(K_V - (\beta K_M + \alpha)\big)^2\Big].$$

Beta Factor & CAPM

Consider the function

$$
\begin{aligned}
f(\beta,\alpha) &= \mathbb{E}\Big[\big(K_V - (\beta K_M + \alpha)\big)^2\Big] \\
&= \mathbb{E}\big[K_V^2 + \beta^2 K_M^2 + \alpha^2 - 2\beta K_V K_M - 2\alpha K_V + 2\alpha\beta K_M\big] \\
&= \mathbb{E}[K_V^2] + \beta^2\mathbb{E}[K_M^2] + \alpha^2 - 2\beta\mathbb{E}[K_V K_M] - 2\alpha\mathbb{E}[K_V] + 2\alpha\beta\mathbb{E}[K_M].
\end{aligned}
$$

To find the minimizer, we set

$$
\begin{cases}
\dfrac{\partial f}{\partial \beta} = 0 \Rightarrow \beta\mathbb{E}[K_M^2] - \mathbb{E}[K_V K_M] + \alpha\mathbb{E}[K_M] = 0; \\
\dfrac{\partial f}{\partial \alpha} = 0 \Rightarrow \alpha - \mathbb{E}[K_V] + \beta\mathbb{E}[K_M] = 0 \Rightarrow \alpha = \mathbb{E}[K_V] - \beta\mathbb{E}[K_M].
\end{cases}
$$

Plugging the second into the first gives

$$
\beta\mathbb{E}[K_M^2] - \mathbb{E}[K_V K_M] + (\mathbb{E}[K_V] - \beta\mathbb{E}[K_M])\mathbb{E}[K_M] = 0,
$$

namely

$$
\beta\big(\mathbb{E}[K_M^2] - \mathbb{E}[K_M]^2\big) = \mathbb{E}[K_V K_M] - \mathbb{E}[K_V]\mathbb{E}[K_M].
$$

This means

$$
\beta_V = \frac{Cov(K_V, K_M)}{\sigma_M^2}, \quad \alpha_V = \mathbb{E}[K_V] - \beta_V\mathbb{E}[K_M] = \mu_V - \beta_V\mu_M.
$$

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The line we eventually find

$$y = \beta_V x + \alpha_V$$

is called the line of best fit, or regression line statistically.

Since $\alpha_V = \mu_V - \beta_V \mu_M$, we can rewrite it as

$$\mu_V = \beta_V \mu_M + \alpha_V.$$

We see that β_V is an indicator on how the (expected) portfolio return K_V changes with the (expected) market return K_M.

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Exercise 6.1

Suppose that the returns K_V on a given portfolio and K_M on the market portfolio take the following values in different market scenarios:

Scenario	Probability	K_V	K_M
ω_1	0.1	-5 %	10 %
ω_2	0.3	0 %	14 %
ω_3	0.4	2 %	12 %
ω_4	0.2	4 %	16 %

Compute the gradient β_V and intercept α_V of the line of best fit.

Answer: $\beta_V = 0.8571$, $\alpha_V = -10.21\%$.

Proposition 6.1

Let two portfolios V_1, V_2 have associated weights $w_1 := (w_{1,1}, \ldots, w_{1,n})$ $w_2 := (w_{2,1}, \ldots, w_{2,n})$. Then,

$$\text{Cov}(K_{V_1}, K_{V_2}) = w_1 C w_2^\top.$$

Proof: Let us denote by $K := (K_1, \ldots, K_n)$ the (random) vector of returns of the risky assets. Now use the bilinearity of the covariance to derive

$$\text{Cov}(K_{V_1}, K_{V_2}) = \text{Cov}\left(w_1 K^\top, w_2 K^\top\right) = \text{Cov}\left(\sum_{i=1}^n w_{1,i} K_i, \sum_{j=1}^n w_{2,j} K_j\right)$$

$$= \sum_{i=1}^n \sum_{j=1}^n w_{1,i} w_{2,j} \text{Cov}(K_i, K_j) = w_1 C w_2^\top.$$

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Proposition 6.2

Suppose $R < \mu_{\text{MVP}}$. Then the expected return μ_V for a feasible portfolio V is given by the CAPM

$$\mu_V = R + \beta_V(\mu_{\text{M}} - R).$$

Proof: We know that the market portfolio exists and its weights are

$$w_{\text{M}} = \frac{(m - Ru)C^{-1}}{(m - Ru)C^{-1}u^\top}.$$

Then,

$$\begin{aligned}\beta_V &= \frac{\text{Cov}(K_V, K_{\text{M}})}{\sigma_{\text{M}}^2} = \frac{w_V C w_{\text{M}}^\top}{w_{\text{M}} C w_{\text{M}}^\top} = \frac{w_V C[(m - Ru)C^{-1}]^\top}{w_{\text{M}} C[(m - Ru)C^{-1}]^\top} \\ &= \frac{w_V (m - Ru)^\top}{w_{\text{M}}(m - Ru)^\top} = \frac{\mu_V - R w_V u^\top}{\mu_{\text{M}} - R w_{\text{M}} u^\top} = \frac{\mu_V - R}{\mu_{\text{M}} - R}.\end{aligned}$$

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Security Market Line

It is clear from the CAPM that on the $(\beta, \mu)-$plane all portfolios lie on the straight line described by CAPM

$$\mu_V = R + \beta_V(\mu_{\mathsf{M}} - R).$$

The graph of this function in the (β, μ)-plane is called the Security Market Line (SML). Note that $\beta_M = 1$ (Why?).

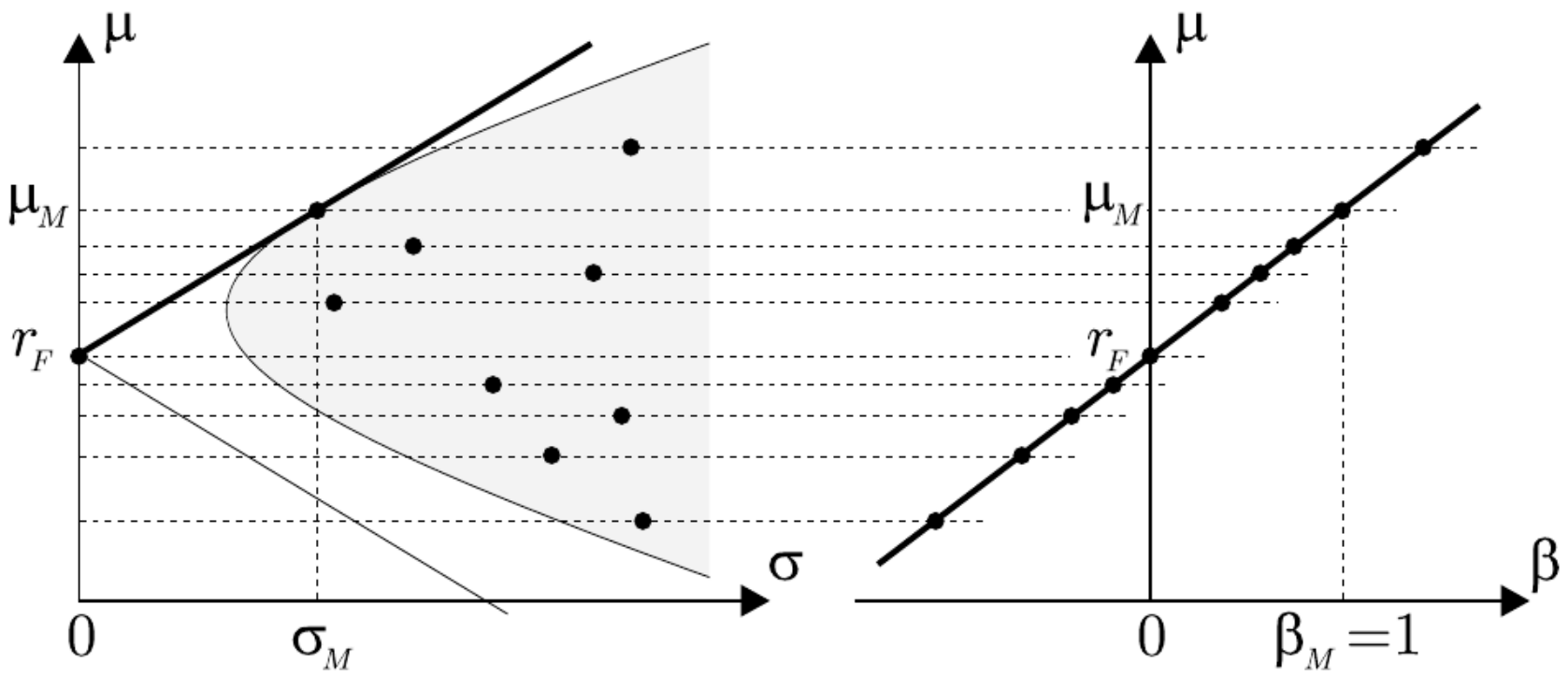

Figure: The Capital Market Line and the Security Market Line

Security Market Line

Two remarks regarding the SML are in order:

1. It is easy to determine the the SML since we know two points $(0, R)$ and $(1, \mu_{\mathsf{M}})$.
2. Despite some securities/portfolios have small returns and large risks, they remain attractive to the investors. The reason is that these securities have negative beta, or equivalently a negative covariance. This means that the prices of these securities tend to move on the opposite direction to the market.

   A standard example of an asset with a negative beta is gold, which acts as an insurance in a financial crisis.

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Value at Risk (VaR)

- Another term related to risk reduction is the Value at Risk.
- We start from nothing today and borrow $\$S(0)$ to buy one share, so that our portfolio is $\big(1, -S(0)\big)$ for the stock and bond. The probability that our loss in 1 year (i.e. $S(0)e^r - S(1)$) is less than a given amount, for example $\$10$, can be expressed as $\mathbb{P}\{S(0)e^r - S(1) < 10\}$.
- Motivated by risk reduction, we are interest in the reverse question: Fix the probability, say 95%, what amount to set today so that our loss in 1 year is below this amount with the given probability 95%?

$$\mathbb{P}\{S(0)e^r - S(1) < ?\} = 95\%$$

This amount is referred to as Value at Risk (VaR) at 95% confidence level of the stock over 1 year.

$$\mathbb{P}\{S(0)e^r - S(1) < \text{VaR}\} = 95\%$$

- In general, it takes to form

$$\mathbb{P}\{\text{the loss of the investment} < \text{VaR}_\alpha\} = \alpha$$

where α is the confident level usually set at 95% or 99%.

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Example 6.1

Let $S(0) = \$100$ and continuously compounding interest rate $r = 5\%$. Suppose that the stock price has a log-normal distribution

$$\text{the log return over one year:} \quad k = \ln \frac{S(1)}{S(0)} \sim \mathcal{N}(0.1, 0.04).$$

If we borrow $\$1000$ from the bank to invest in this stock, what is the VaR at 95% confidence level over one year? Standard normal cdf $\Phi(1.645) = 95\%$.

Loss of the investment in one year $= 1000e^r - 10S(1)$. The VaR is such that

$$\begin{aligned} 95\% &= \mathbb{P}(1000e^r - 10S(1) < \text{VaR}) \\ &= \mathbb{P}(1000e^r - 10S(0)e^k < \text{VaR}) \\ &= \mathbb{P}(k > \ln \frac{1000e^r - \text{VaR}}{10S(0)}) \\ &= \mathbb{P}(\frac{k - 0.1}{0.2} > \frac{\ln \frac{1000e^r - \text{VaR}}{10S(0)} - 0.1}{0.2}) \end{aligned}$$

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Eg 6.1 continued Since $\frac{k-0.1}{0.2}$ is a standard normal random variable and we are given the standard normal cdf $\Phi(1.645) = 95\%$,

$$\frac{\ln \frac{1000e^r - \text{VaR}}{10S(0)} - 0.1}{0.2} = -1.645.$$

From this we get

$$\text{VaR} = 255.94.$$

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